

ENV 5128

Smart Air Quality Monitoring & Air Pollution Control

Today's Topics

- Refresh of air pollution basics
- Expectation
 - Familiar with air pollution issues at different scales
 - Familiar with source types of air pollution
 - Familiar with the wide range of adverse health and environmental effects of air pollution
 - Understand CAA, NAAQS, CAP, HAP, NESHAPS, SIP, NSR, PSD
 - Know the six CAPs and their major sources
 - Understand and be able to interpret wind rose map

Air pollution issues at different scales

What is air pollution

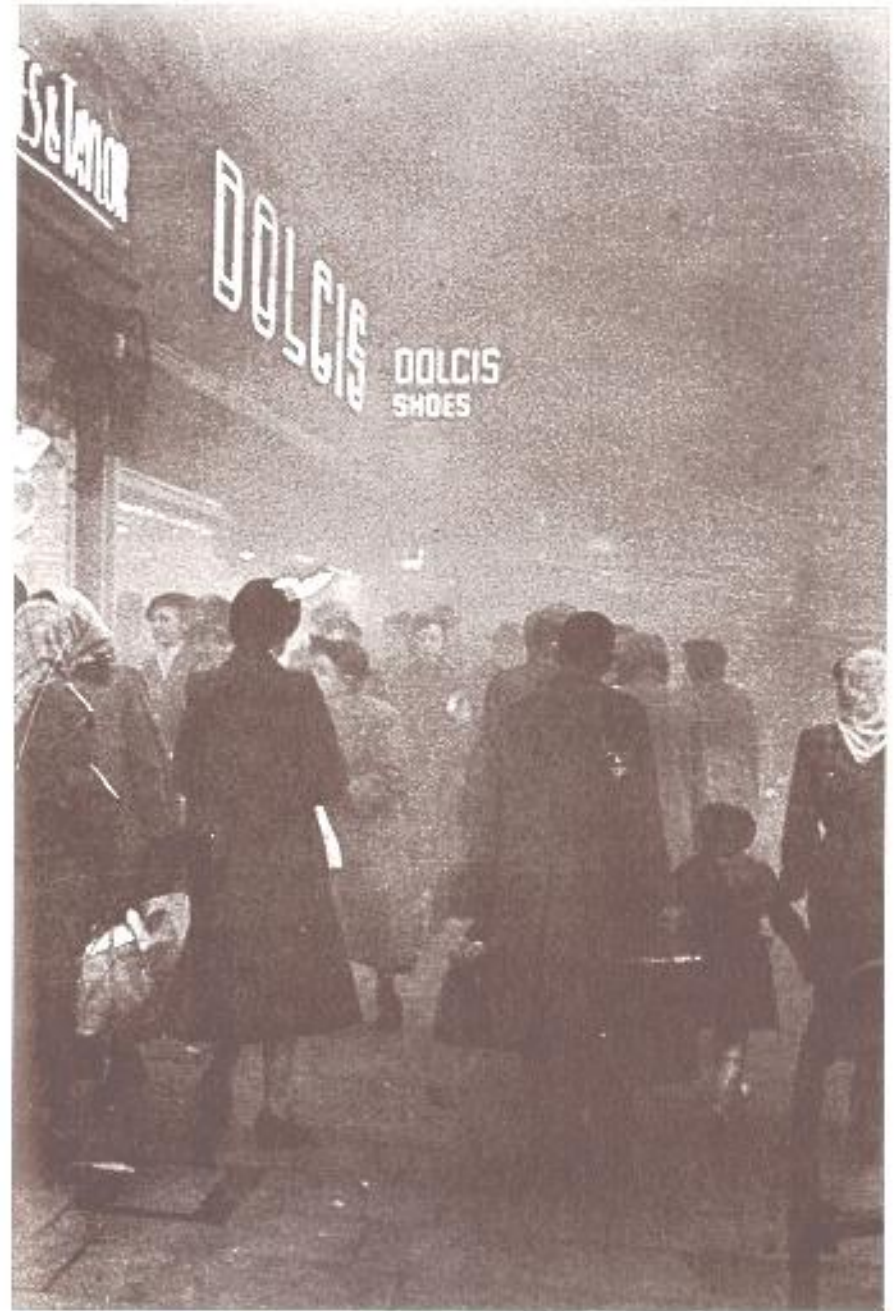
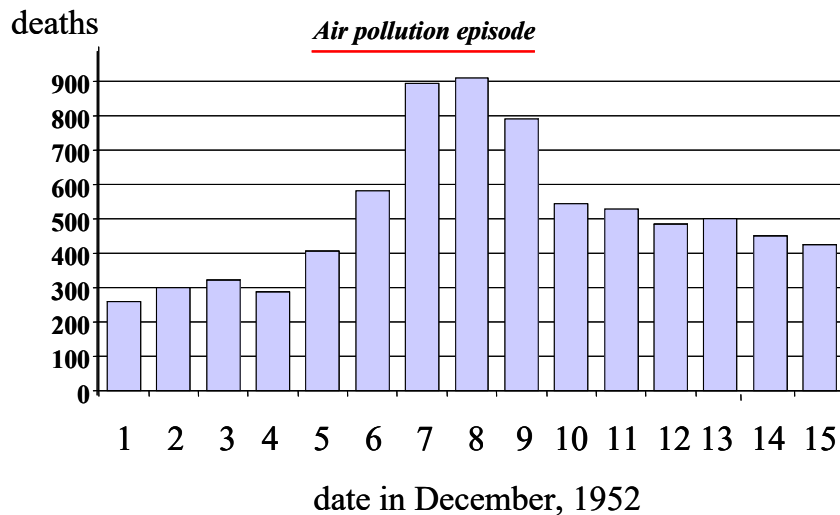
- As defined in Florida Administrative Code:
- “Air pollution is the presence in the outside atmosphere ... of any one or more substances or pollutants in quantities which are or may be harmful or injurious to human health or welfare, animal or plant life, or property, or unreasonably interfere with the enjoyment of life or property, including outdoor recreation.”

Scales of air pollution problems

- Personal
- Micro-environmental (meters) – indoor air pollution
- Local (< 5 km) – hot spots, near source
- Urban (~ 50 km) – urban smog
- Regional (50 – 500 km) – acid deposition, visibility
- Continental (500 – 1000 km) – long-range transport, stratospheric ozone depletion
- Global – stratospheric ozone depletion, long range transport, climate change

London smog, 1952

- December temperature inversion
- air pollution levels (PM, SO₂) 5x higher than normal
- 4,000 excess deaths attributed to fog
- 80% of deaths due to cardiorespiratory disease (30% typ)



LA smog



Beijing smog



Smog in 20 minutes. Beijing, China. Jan 2nd, 2017
Credit: YouTube/Chas Pope

Regional air pollution (haze)

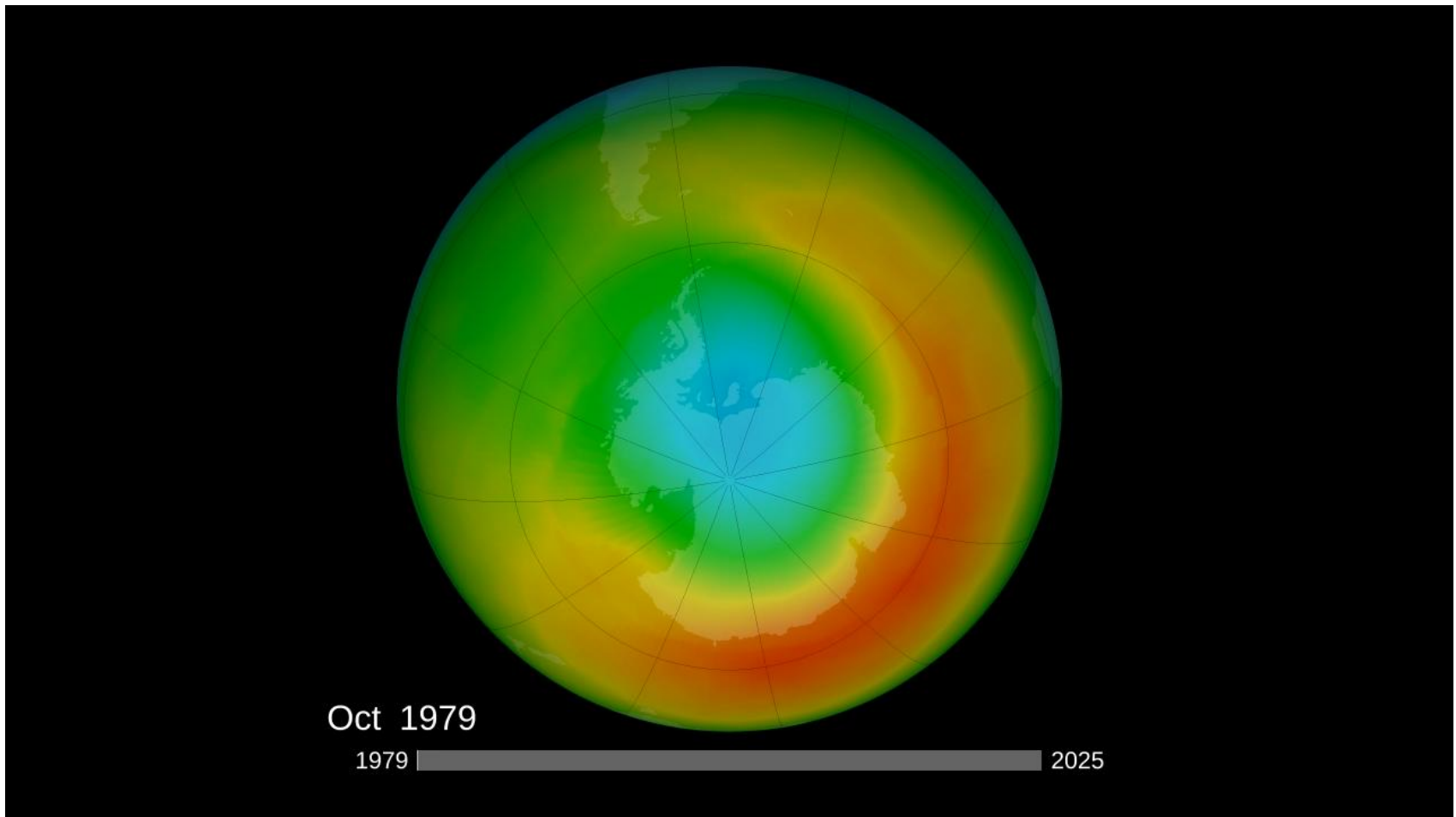
Shenandoah Valley



Acid deposition

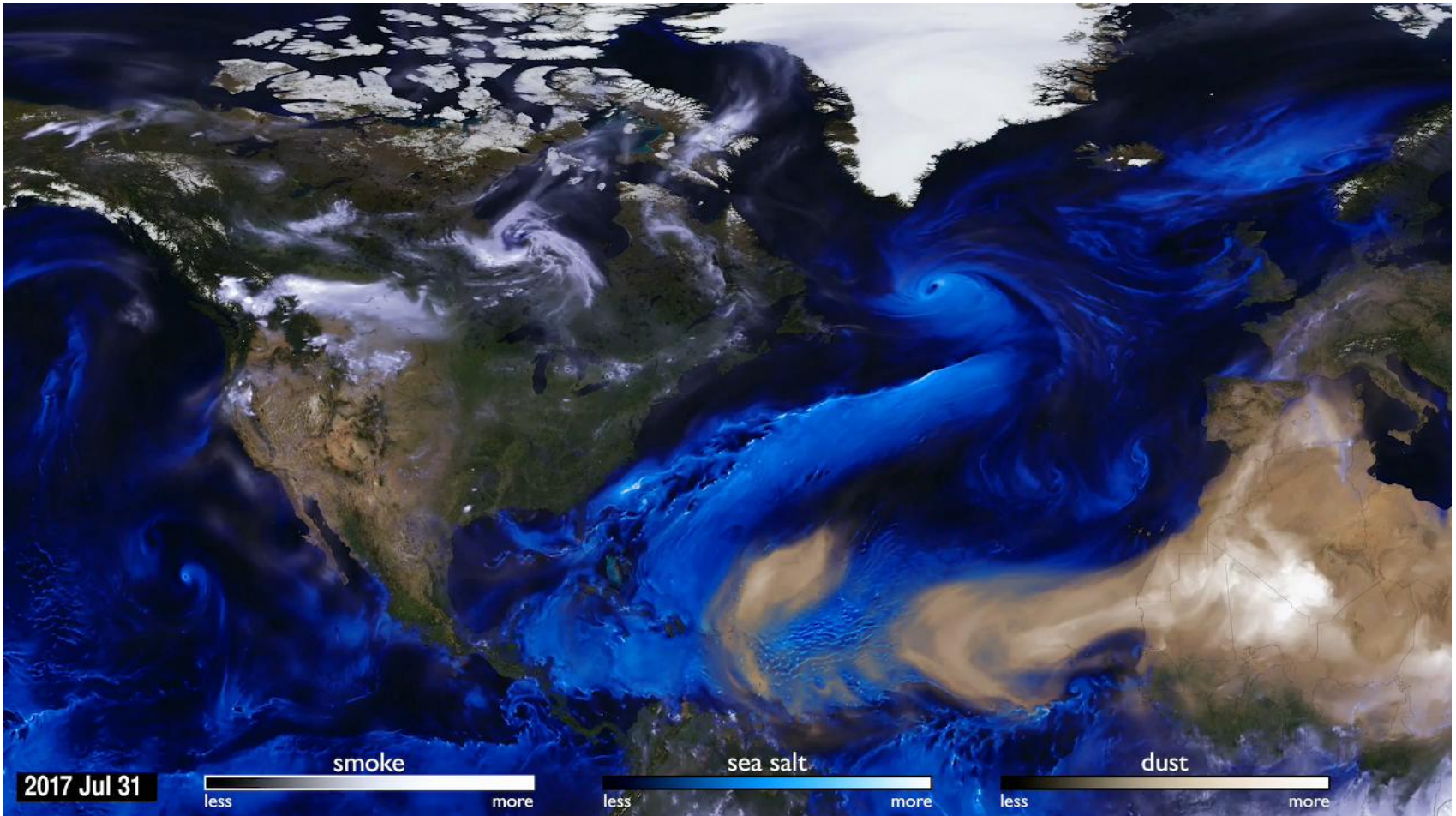


Stratospheric ozone depletion



Antarctic September ozone hole from 1979 to 2022
(NASA Ozone watch <http://ozonewatch.gsfc.nasa.gov/monthly/SH.html>)

Wildfires



2017 Hurricanes and Aerosols Simulation from NASA
(<https://svs.gsfc.nasa.gov/12772>)
Latest satellite image of fire see NASA worldview

Brief U.S. Federal Legislative History

Brief U.S. Federal Legislative History

- Mid 1800s: industrialized cities were well established as very dirty places
 - New York, Chicago, Pittsburgh etc...



Pittsburgh from the Salt Works at Saw Mill, by William T. Russell, 1843
(<http://explorepahistory.com/displayimage.php?imgId=1-2-D44>)

Brief U.S. Federal Legislative History

- By turn of the century, air pollution becoming recognized as a local health problem
- 1907: The International Association for the Prevention of Smoke established
 - Later the Air Pollution Control Association, now the Air & Waste Management Association
- Automobile revolution and beginnings of American consumer culture

Brief U.S. Federal Legislative History

- 1940s: LA haze; Donora (PA) smog (20 deaths)
- 1952: London smog (4000 deaths)
- 1955: Clean Air Legislation (1st Fed. regulation)
 - Authorized Public Health Service (Dept. of Health, Education, and Welfare: DHEW) to:
 - Conduct air pollution research and training programs
 - Provide technical assistance to state and local governments

London smog
(www.klimaforschung.net)



Donora, PA at noon on
October 29, 1948
<http://pabook.libraries.psu.edu/palitmap/DonoraSmog.html>

Brief U.S. Federal Legislative History

- 1960s: Social change and significant national environmental movements
 - Sierra Club, Wilderness Society
 - 1962: Rachel Carson's 'Silent Spring'
- 1963: Clean Air Act (CAA)
 - Broadened federal role in air quality management
 - Grants for program development/improvement of state and local air pollution control
 - Research, training, and technical assistance
 - Federal enforcement authority for interstate air pollution problems
 - Federal research responsibility for automobile and SO_x pollution
 - Authority to compile and publish criteria for pollutants

Brief U.S. Federal Legislative History

- 1965: Motor Vehicle Air Pollution Control Act
 - Amendment to 1963 CAA
 - Established the National Air Pollution Control Administration (NAPCA)
 - Authorized DHEW/NAPCA to promulgate motor vehicle emissions standards

Brief U.S. Federal Legislative History

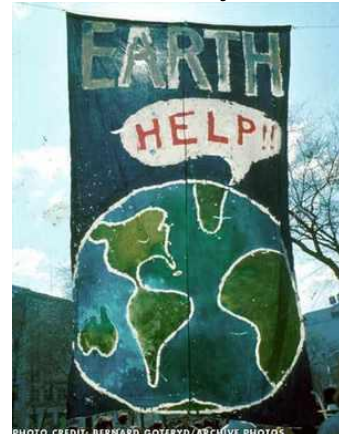
- 1967: Air Quality Act (Amendment to CAA)
 - Institution of a regional approach to air pollution control and standard setting
 - Air Quality Control Regions (AQCR)
 - NAPCA to develop/issue air quality criteria and control technique info
 - States to develop state air quality standards, **state implementation plan (SIP)**



Brief U.S. Federal Legislative History

- 1970: USEPA established (NAPCA eliminated)
- 1970: Clean Air Act Amendments
 - Provided for federal enforcement authority
 - Stringent auto emissions and fuel standards
 - National Ambient Air Quality Standards (NAAQS)
 - Six Criteria Air Pollutants (CAP)
 - New Source Performance Standard (NSPS)
 - National Emissions Standards for Hazardous Air Pollutants (NESHAPS)

First earth day, 1970



Brief U.S. Federal Legislative History

- 1977: Clean Air Act Amendments
 - Required review of NAAQS every 5 years
 - Non-compliance penalties
 - New Source Review (NSR) for major sources
 - Prevention of Significant Deterioration (PSD)
 - Grant USEPA authority to regulate ozone destroying chemicals
- 1977: National Atmospheric Deposition Program (NADP) established
 - Monitoring trends of acidity in precipitation

Brief U.S. Federal Legislative History

- 1978: US ban manufacture and sale of CFCs for spray cans
- 1980: Acid Precipitation Act
 - National Acid Precipitation Assessment Program (NAPA)
 - Enlarged NADP to National Trends Network (NTN)
- 1980: Election of Ronald Reagan
 - “Onslaught” on environmental regulations

Brief U.S. Federal Legislative History

- 1990: Clean Air Act Amendments
 - Reauthorization and update of CAA
 - Strengthen measures for attaining air quality standards
 - Boundaries of non-attainment areas to include all areas in metropolitan area
 - Expanded regulation of HAPs
 - Max achievable control technology (MACT)
 - Establish program for prevention of accidental releases
 - Updated authority and specific requirements for stratospheric ozone depletion
 - Expanded enforcement powers and penalties
 - Criminal sanctions for negligent violations

Brief U.S. Federal Legislative History

- 2005: Clean Air Interstate Rule (CAIR)
 - Cap and trade for O₃ and PM
 - Replaced by Cross-State Air Pollution Rule (CSAPR) in 2011
- 2007: Supreme court ruling providing EPA authority over CO₂
- 2015: Volkswagen emissions scandal (\$2.8B fine)
- 2015: Clean Power Plan: reduce carbon emissions from power plants
- 2016: 1st Trump presidency
 - Many environmental policy changes followed

Brief U.S. Federal Legislative History

- 2017: Dismantle of Clean Power Plan
 - Held by supreme court in 2016
 - Replaced with Affordable Clean Energy rule in 2019
 - One example of Clean Coal: Kemper Project
- 2017 & Beyond: ?????
 - Pull out of Paris climate accord, twice
 - Inflation Reduction Act
 - Drill, baby, drill
 - Biggest USEPA deregulatory action in U.S. History
 - & countless others

Sources of air pollution

Emission sources

- Primary vs secondary
 - Directly emitted vs formed in the air
- Natural vs man-made (anthropogenic)
- Stationary vs mobile
 - Stationary stacks, moving cars etc...
- And also by shapes
 - Point, nonpoint, area, volume, line etc....

Natural Sources

Volcano



USGS image by C. Newhall - September, 1984 (www.geology.com)

Sea



Credit: Photo by Michele Hogan
<http://www.sciencedaily.com/releases/2007/09/070919175542.htm>

Forest



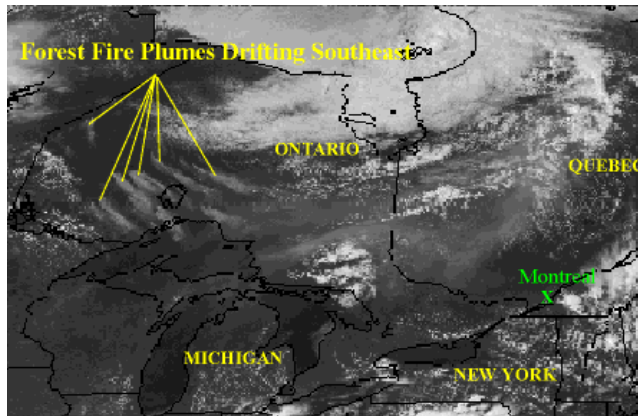
www.ivm.vu.nl/en/projects/Departments/economics/Reduced_Emission_from_Deforestation_and_Degradation_in_the_Southern_Cardamom_Ecosystem_Cambodia/index.asp

Dust storm



NOAA Photo Library, Historic NWS collection

Forest fire



Satellite image for June 14, 1996, at 1545 EST
(www.radar.mcgill.ca/forest_fire.html)

Hot springs



www.phase2.net/claygraphics/Images/bruce%20images/Tahcharian%20hot%20Springs.jpg

Anthropogenic Sources

Power plants



(David Klobucar/Associated Press)
<http://www.cbc.ca/news/canadavotes/story/2008/09/19/f-carbontaxprimer.html>

Automobile



www.hybridmile.com/news/obama-supports-californi/

Industry



pollutionarticles.blogspot.com/2009/02/carbon-emissions-causing-huge-ocean.html

Lawn care



www.re-nest.com/uimages/re-nest/mowerwoody052509.jpg

Farms



www.extension.iastate.edu/Pages/communications/EPC/Su02/emissions.html

Land fill



Landfill <http://www.jber.af.mil/shared/media/photodb/photos/2012/08/120821-F-BP133-050.jpg>

CAPs and HAPs

CAPs & HAPs

- 6 Criteria Air Pollutants (CAPs):
 - O₃, CO, NO₂, SO₂, Pb, PM (PM_{2.5} & PM₁₀)
 - Concentration level set in NAAQS (in CAA)
 - Ubiquitous, most harmful to health and environment
- 189 Hazardous Air Pollutants (HAPs)
 - A.K.A. toxic air pollutants or air toxics
 - Known/suspected to cause cancer, or other serious health or environmental effects
 - Goal: reduce emissions (NESHAPS)
 - Primary, man-made emissions dominates

Particulate Matter (PM)

- A.K.A. aerosol, can be a solid/liquid/mixed
- PM_{10} & $PM_{2.5}$
 - PMs with aerodynamic diameter < 10 or $2.5 \mu m$
 - $1 \mu m = 1 \times 10^{-6} m$
 - Can lodge in airway once inhaled.
 - $PM_{2.5}$ penetrate deep into lungs (alveoli region)
- Many emission sources
 - Fire, dust, agriculture, fuel combustion, cars etc...
 - Both primary/secondary, natural/man-made sources
 - Secondary organic aerosol (SOA) is a major research topic

Nitrogen Dioxide (NO₂)

- Usually NO and NO₂ are measured together
 - NO eventually oxidized to NO₂
 - NO + NO₂ = NO_x (different from NO_y and NO_z!)
 - NO_z = a combination of several reactive nitrogen species
 - NO_y = NO_x + NO_z
- Mostly man-made, primary emissions (fuel combustion at high temperatures)
 - Mobile source, power plants dominates
- Some natural sources
 - Lightning generate NO_x
 - Some plants also emit NO_x

Ground-level ozone (O_3)

- One major component of photochemical smog
- Ground-level O_3 BAD; stratospheric O_3 GOOD.
 - Health effect when exposed near ground
 - Stratospheric O_3 layer absorbs harmful UV
- Secondary pollutants, negligible direct emissions
 - Primarily from NO_x and VOC, mixing with sunlight
 - VOC: Volatile Organic Compounds
 - From evaporation of organic solvents, incomplete combustion etc...
 - Trees emits VOC (biogenic): mostly isoprene
 - Globally, trees emits more VOC than man-made sources
 - Ronald Reagan (1981): Trees cause more pollution than automobiles do

Sulfur Dioxide (SO₂)

- From combustion of sulfur-containing fuel
- Primary, anthropogenic sources dominate
 - Coal fire power plants is the biggest source
 - Also industry, mobile, fire (wild + prescribed)
 - Some natural source: sulfur springs
- Major contributor to acid rain, especially northeastern US

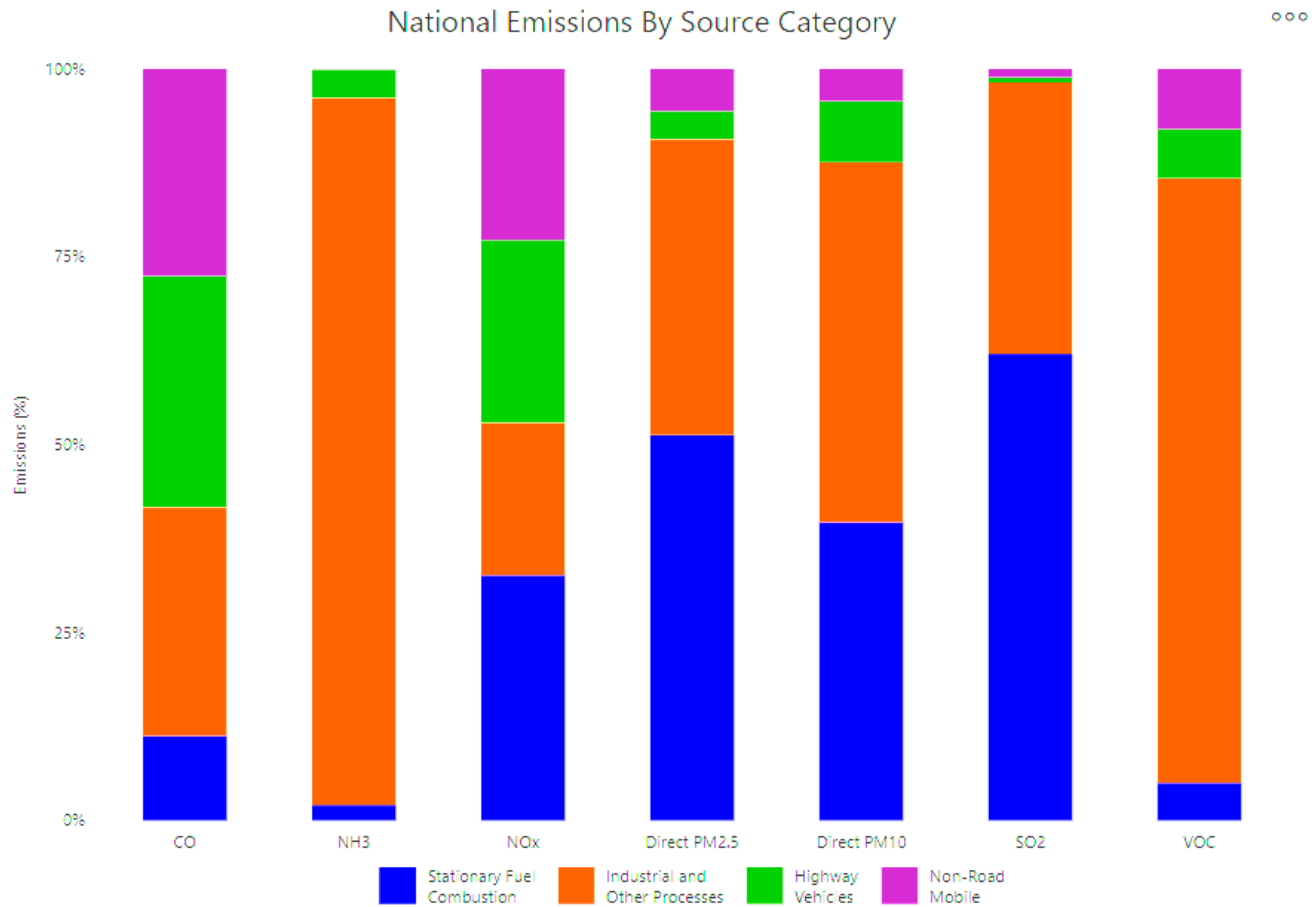
Carbon Monoxide (CO)

- Colorless, odorless, and tasteless gas
- Mostly primary, from incomplete fuel combustion
 - Mobile source, fire are the biggest ones
- Some plants emits CO (biogenic)
- CO binds with hemoglobin, reduce oxygen carrying capacity of blood

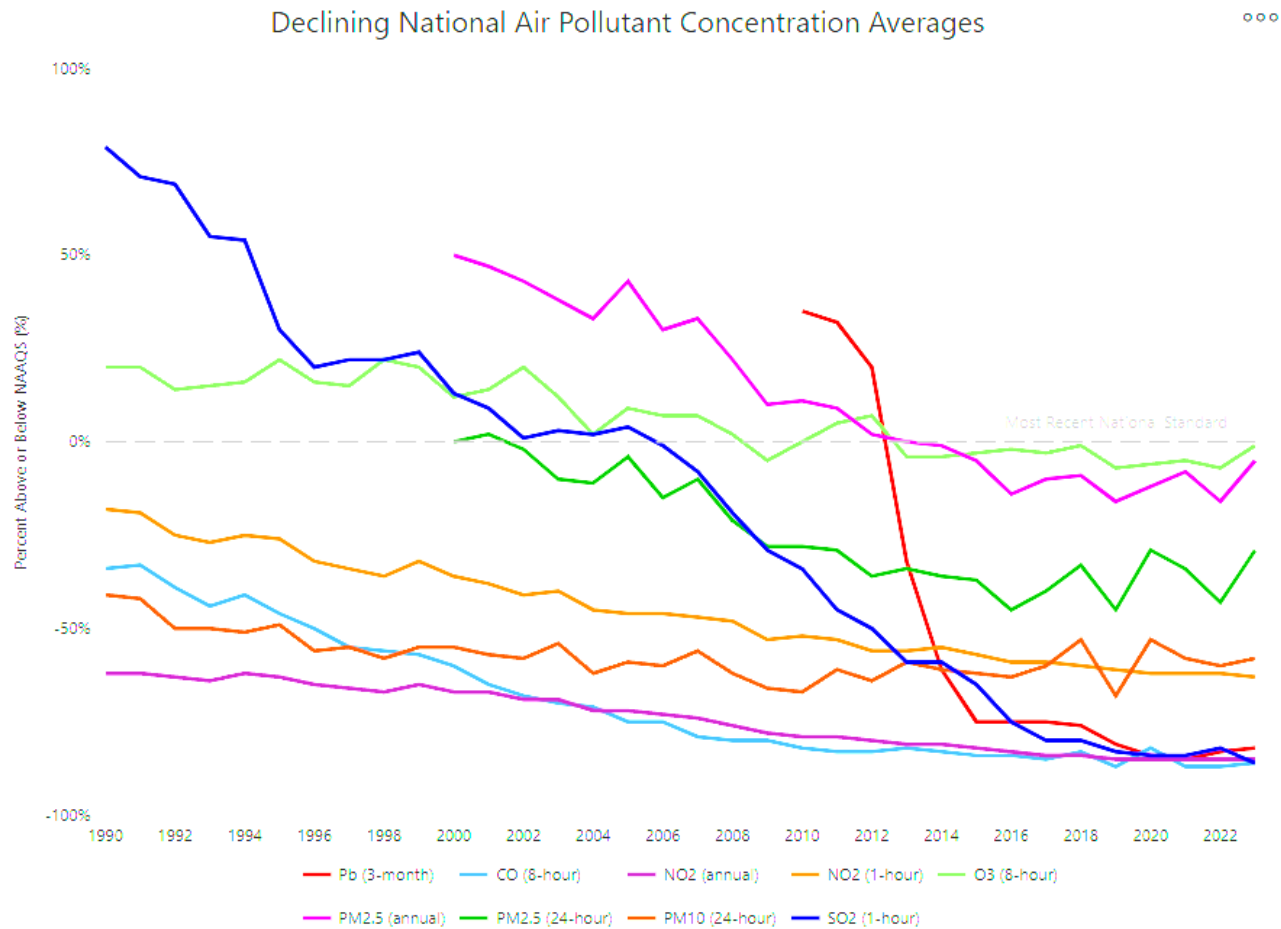
Lead (Pb)

- Particulate lead
- Combustion of fuels with lead additives
- Leaded gas phased out → lead in ambient air decreased by 99%
 - Leaded gas invented by Thomas Midgley Jr. who also brought us Freon (CFC)
- Lead particularly toxic to children
- Houses built before 1978 has lead paint

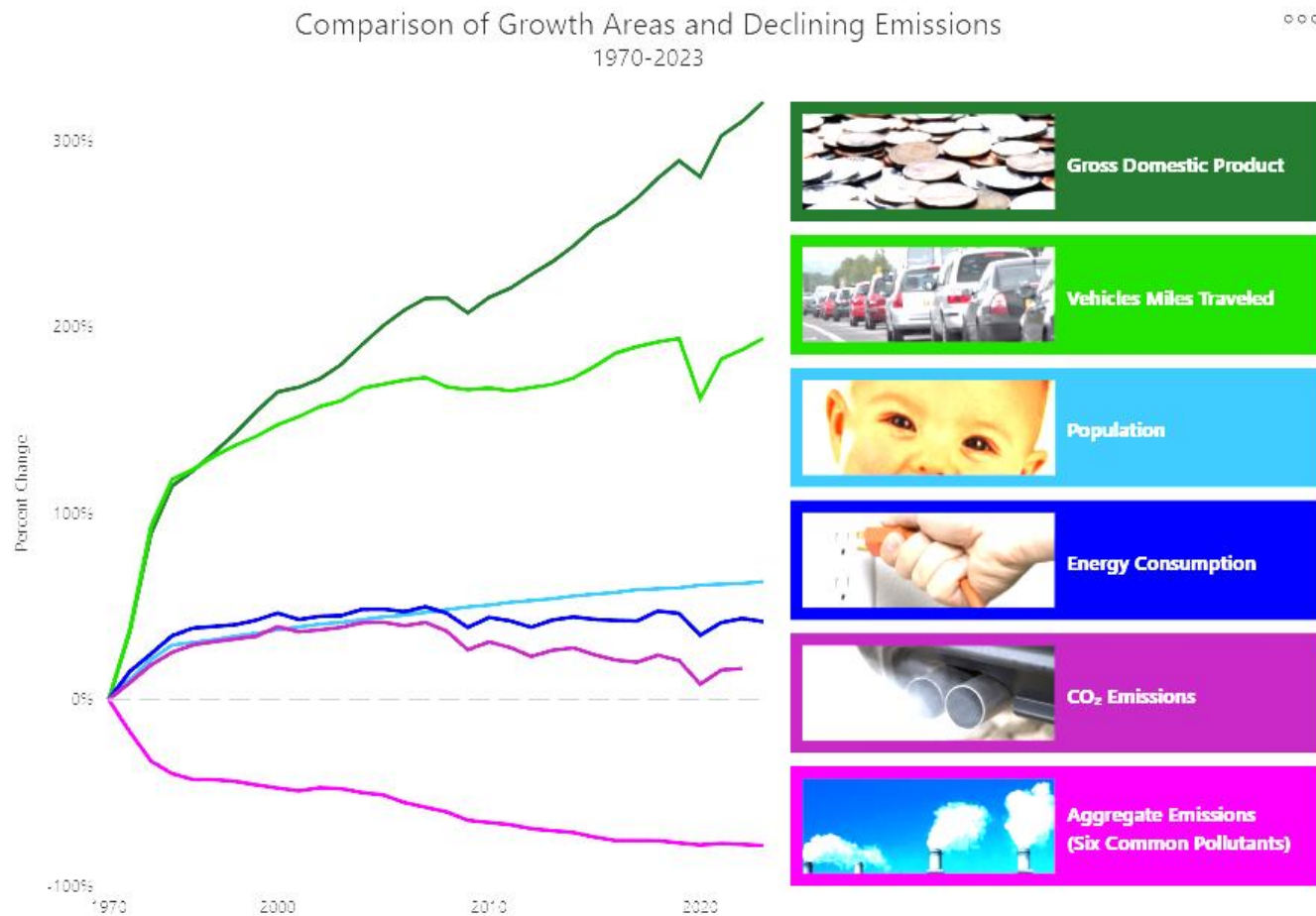
Emission sources



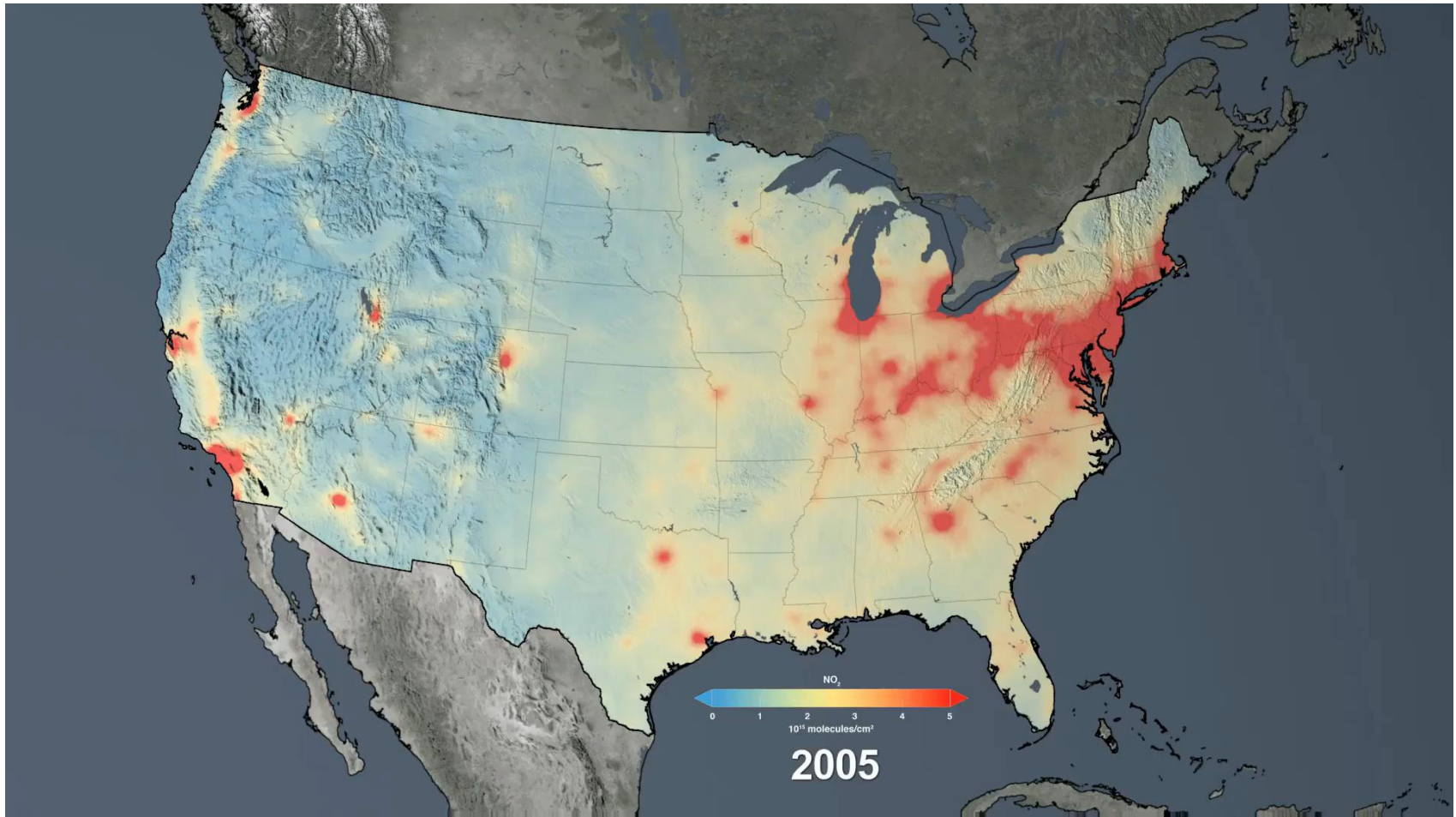
Concentration trend in the US



Emission trend in the US

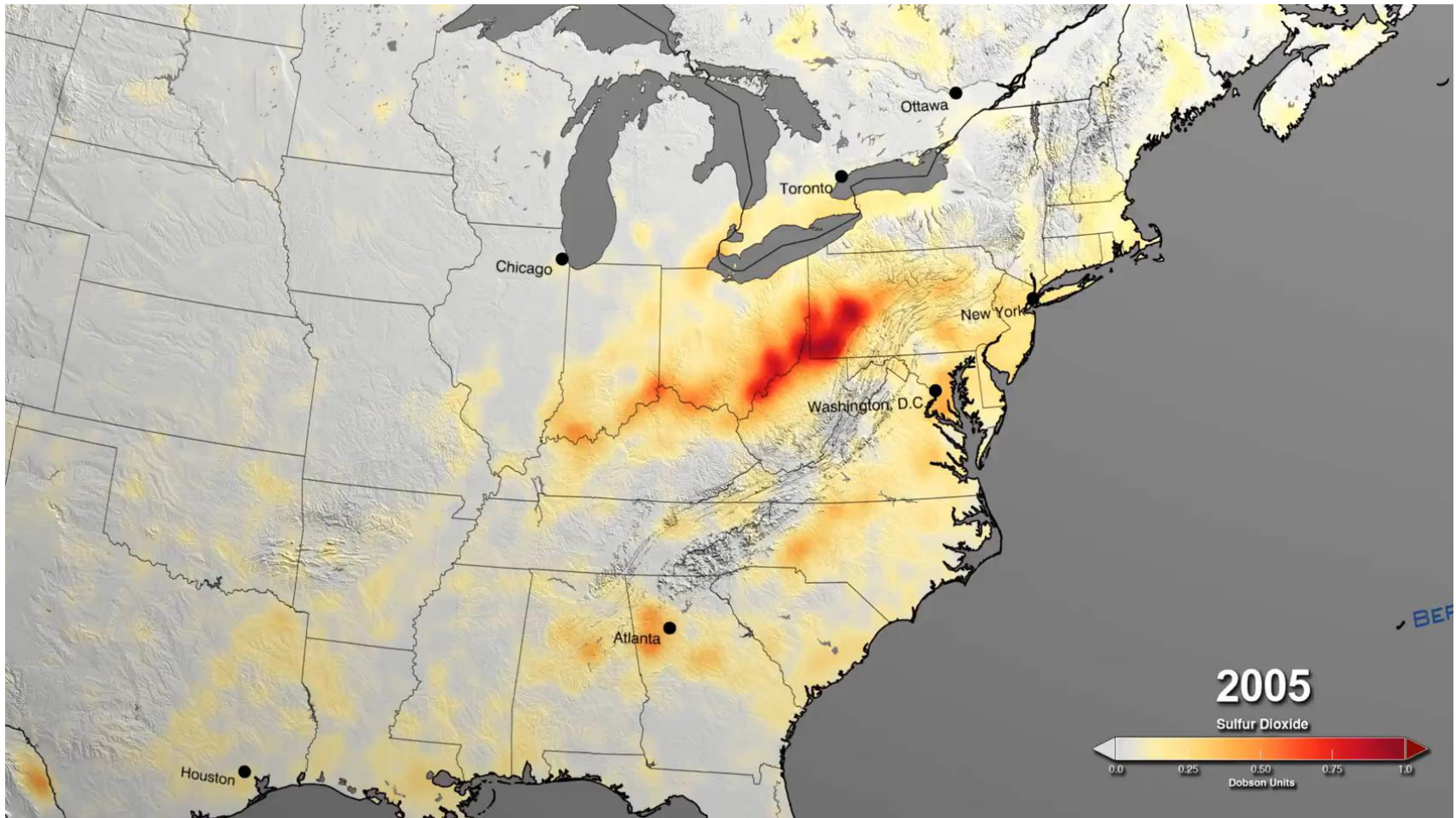


NASA satellite NO₂ concentration



More information on trend of CAPs see:
<https://gispub.epa.gov/air/trendsreport/2024>
Also see here: <https://airquality.gsfc.nasa.gov/>

NASA satellite SO₂ concentration



Transport of air pollution

Pollution transport

- After emitted/formed, pollutants need to be transported to get in contact with “receptor”
 - Human, certain part of eco-system etc..
- Pollution transport depends on meteorology
- Pollutants can be removed or formed during the process
- The distance of transport differ drastically depends on pollutant characteristics

Gone with the wind – Wind Rose

Horizontal mean flow

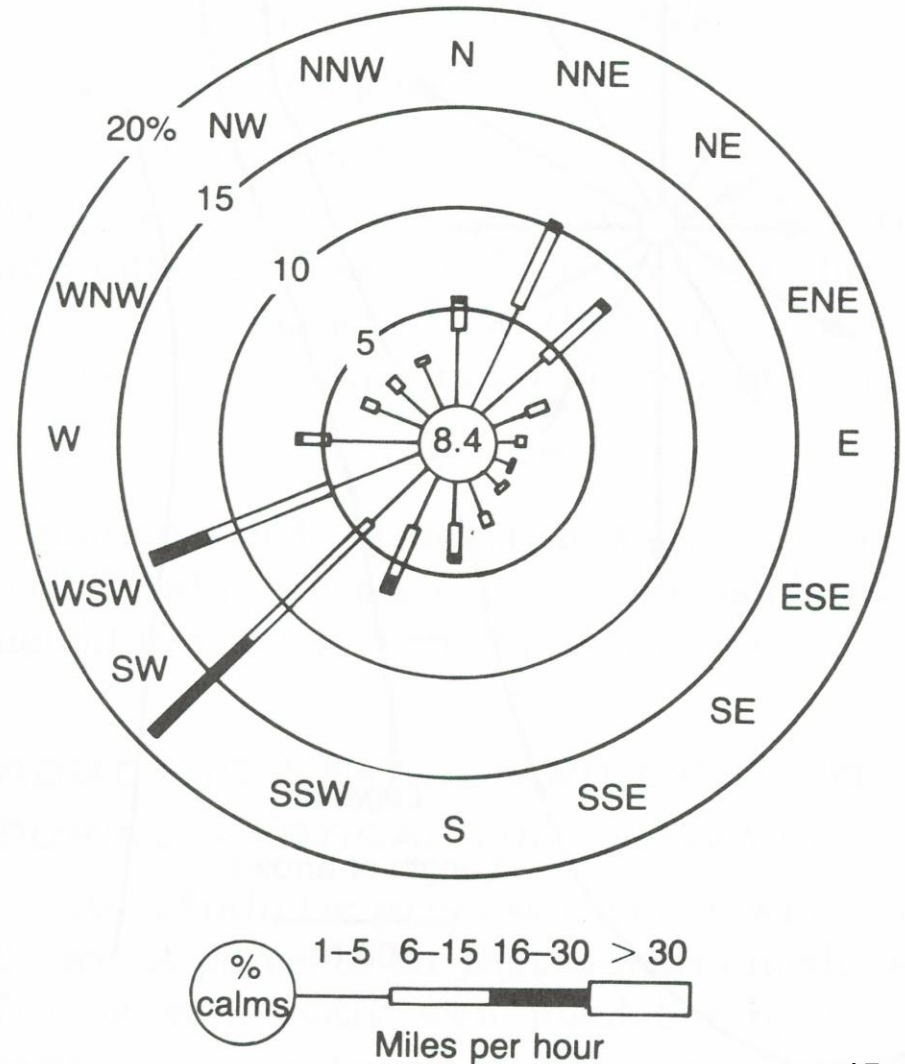
Direction

Where wind is coming from
(angle in degrees clockwise
from the north)

Direction of pollutant transport

Speed

transports pollutants further
dilutes emissions
creates turbulent mixing
bends plumes

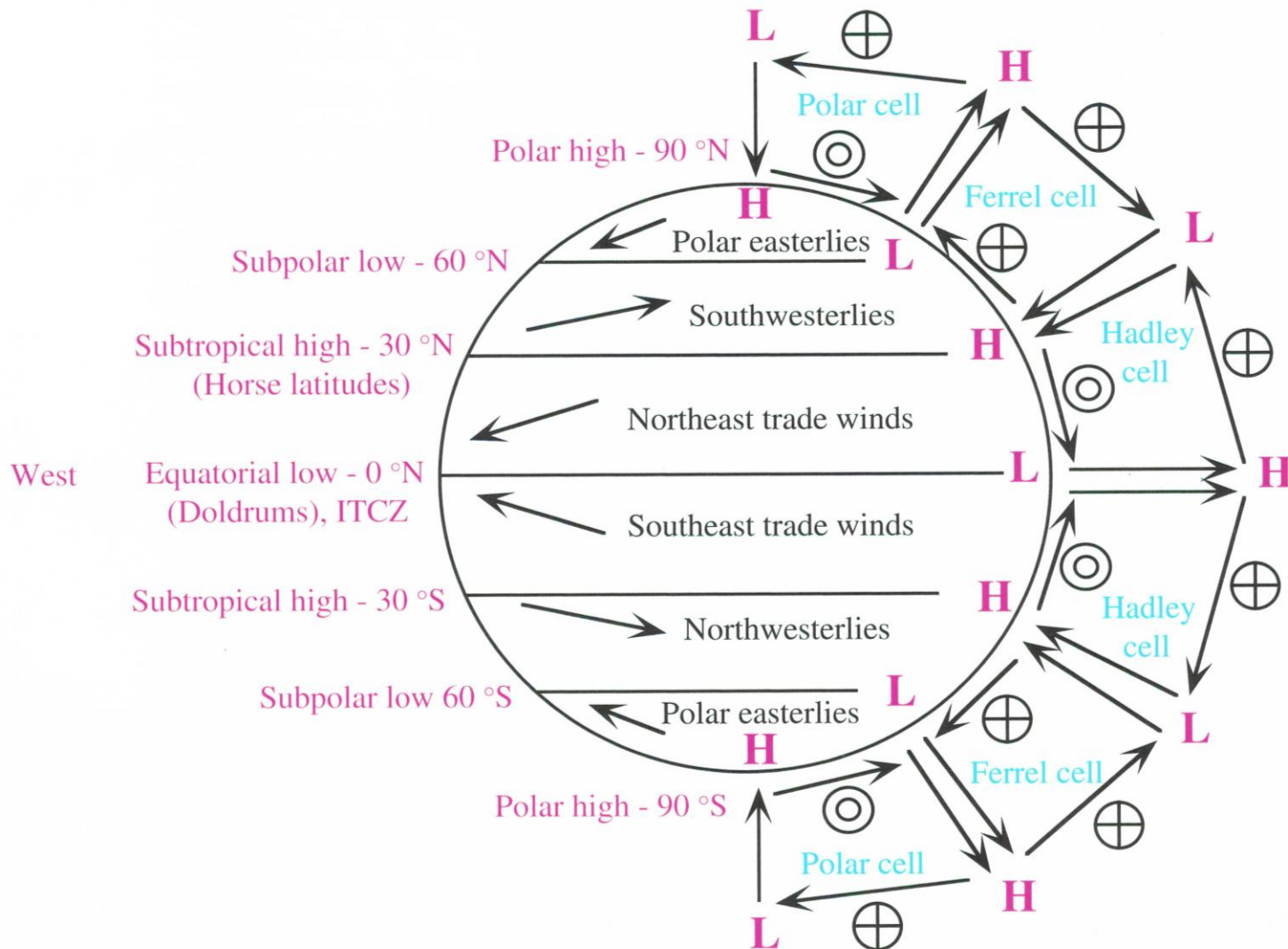


Gone with the wind



Klyuchevskaya Sopka volcano, Russia. 9/30/1994 eruption. Taken by Endeavour
Credit: NASA

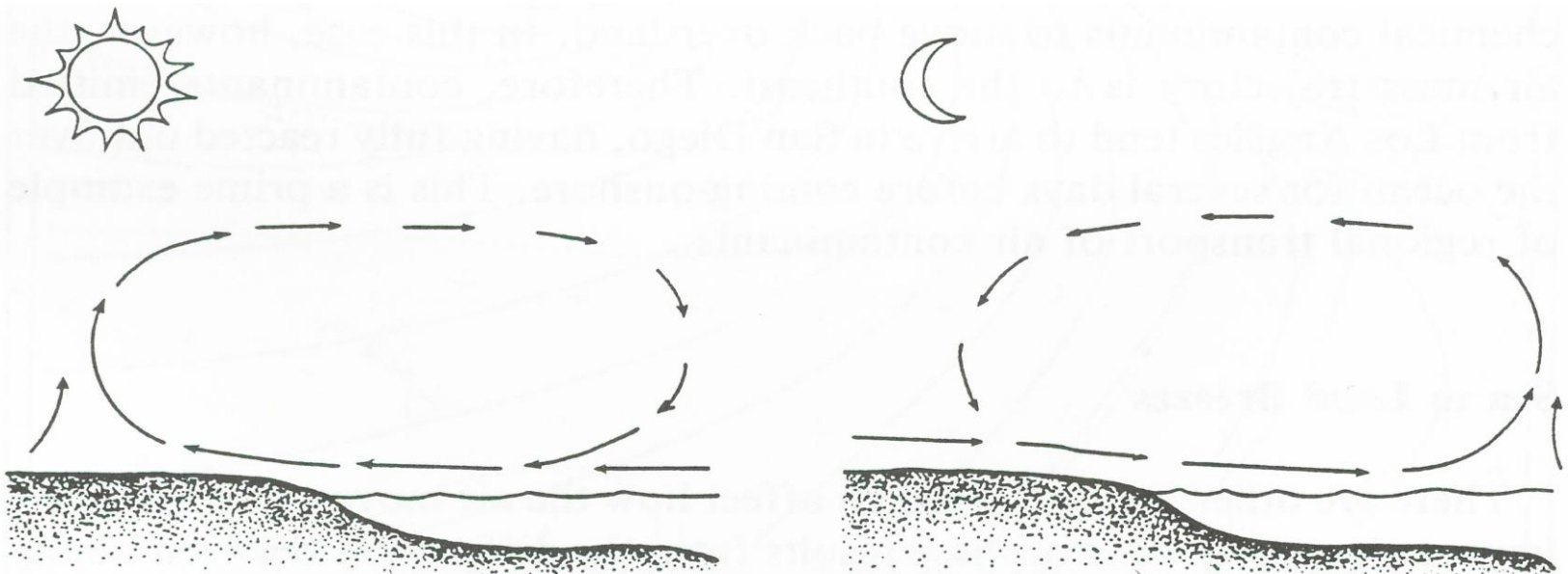
Large Scale Motion: General Circulation pattern



(Jacobson, 2002)

Smaller scale motion

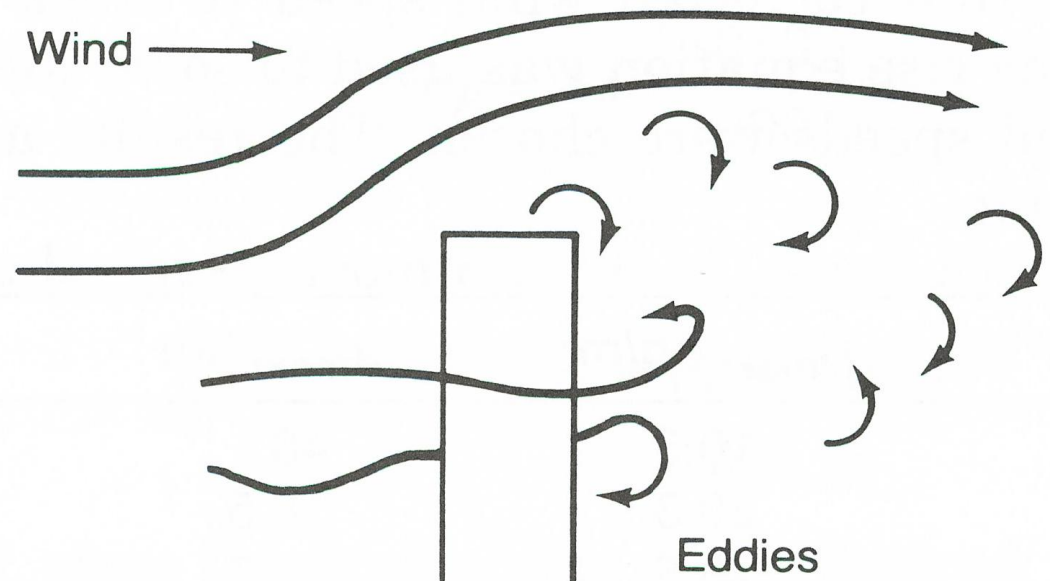
- Impact by variable heating and topography
 - valley and mountain breezes
 - sea and land breezes
 - urban heat island flow



Example of land sea breeze (Griffin, 1994)

Even smaller scale

- Turbulence
 - Random / irregular fluctuations on top of mean flow
 - Circular eddies at smaller scale than mean flow
- Important to
 - mixing
 - dilution
- Pollution hot spot



Effects of air pollution

Air Pollution Health Effects

- Effects on human
 - Numerous adverse health effects, ranging from minor irritation, discomfort, to chest pain, asthma, birth defect, cancer and death
- Effects on the environment
 - Damage property/crops/plants (ozone, acid rain etc..)
 - Visibility impairment
 - Eutrophication (excessive nutrition, e.g. nitrogen)
 - Damage ozone layer → increase harmful UV radiation
 - Numerous others....

Air Pollution Health Effects

Respiratory

Coughing, wheezing,
reduced lung function

Exacerbation of asthma,
COPD

Lung cancer

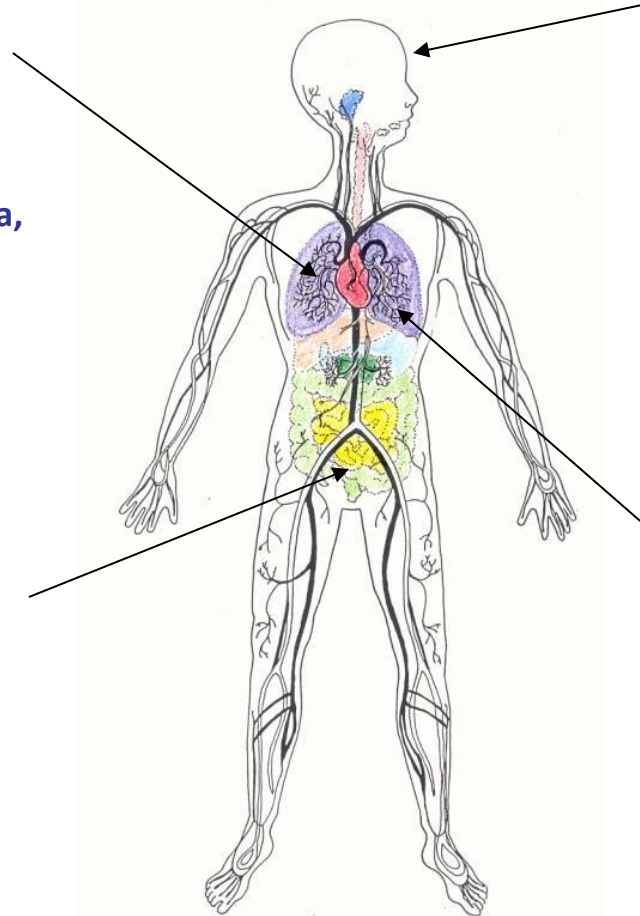
Respiratory mortality

Reproductive

Low birth weight

Preterm births and
intrauterine growth
retardation (?)

Birth defects (?)



Central Nervous

Stroke (?)

Cognitive effects(?)

Cardiovascular

HRV reduction, dysrhythmias

Systemic inflammation

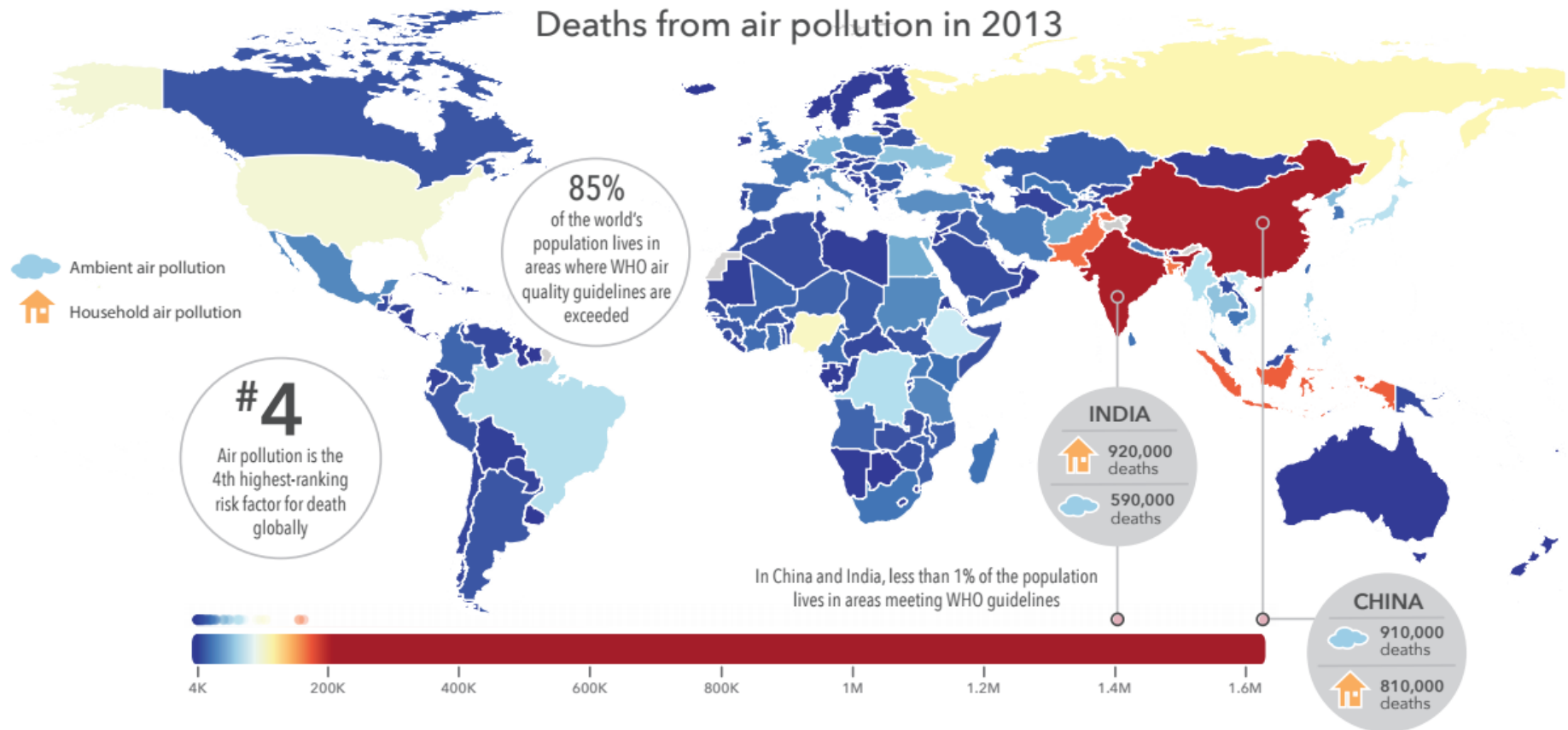
Atherosclerosis

Myocardial infarctions (Heart Attacks)

CV mortality

Air pollution kills

Global Burden of Air Pollution



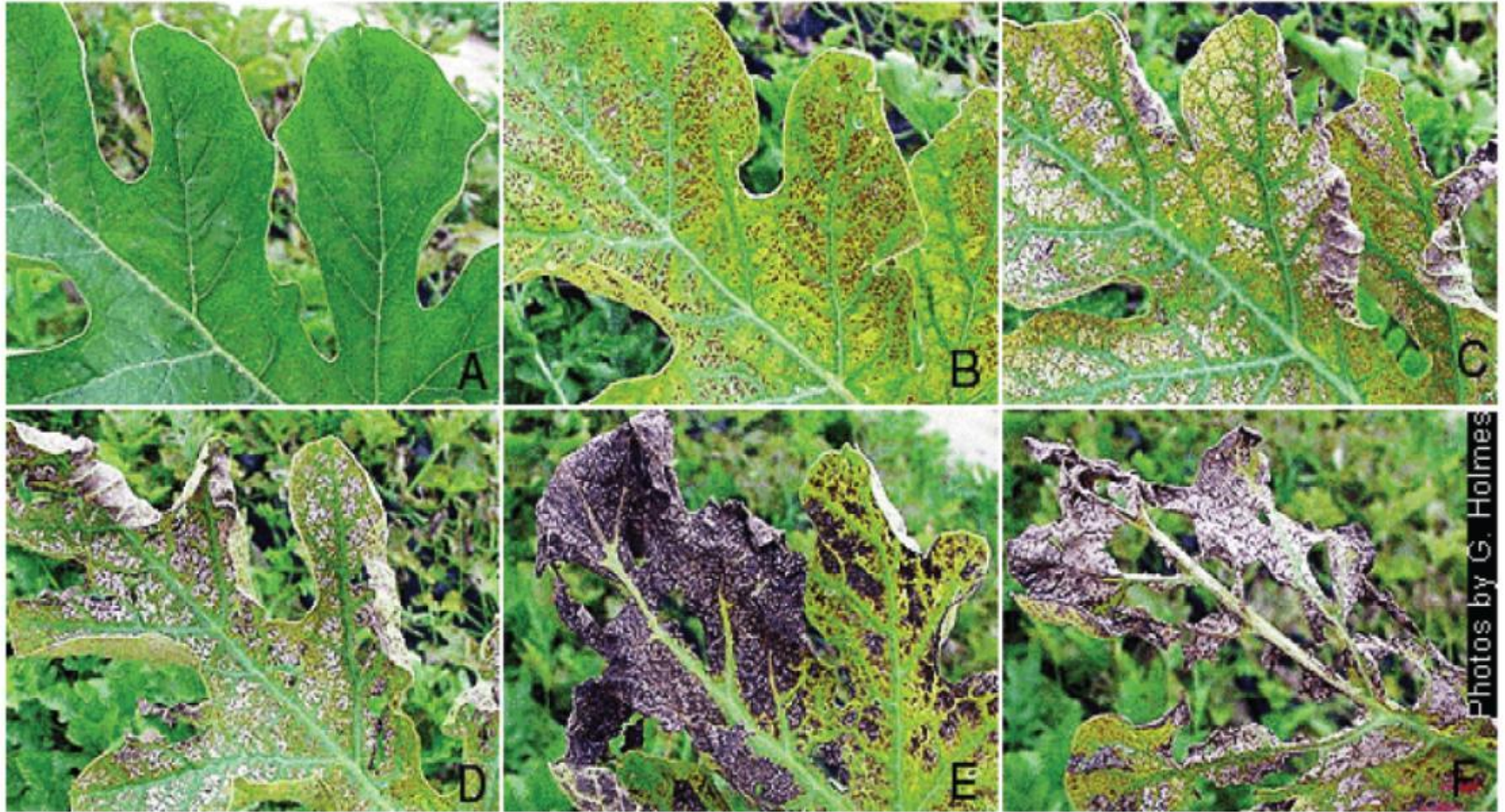
Air pollution was responsible for 5.5 million deaths in 2013

Credit: www.healthdata.org. Global Burden of Disease team

Ozone damaged crops

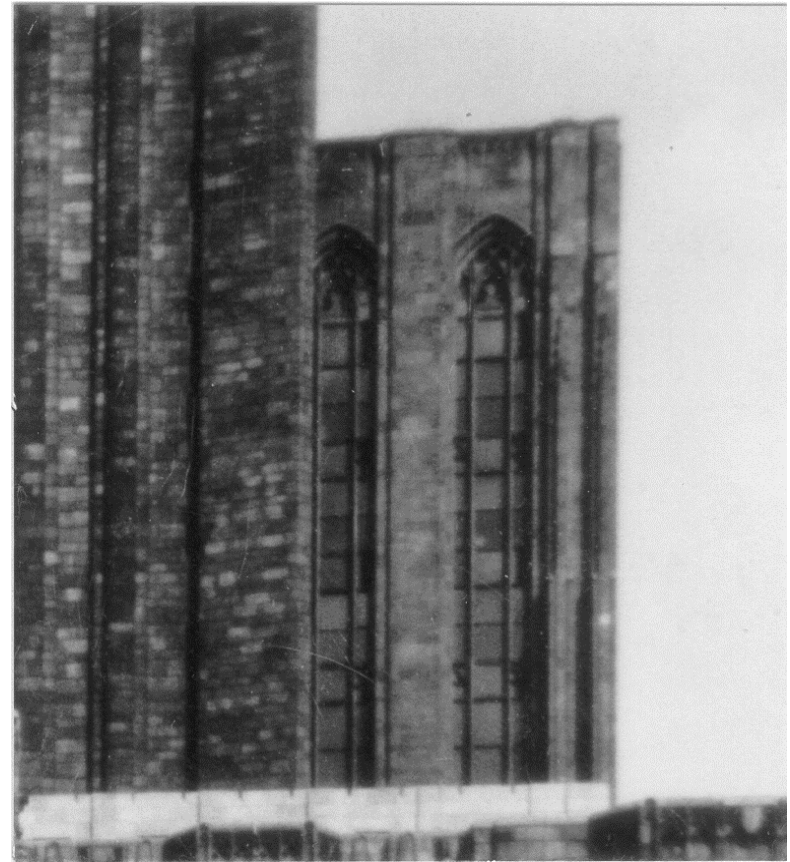
Fig. 2 Progression of ozone damage (A=none to F=severe) on watermelon foliage.

Photo courtesy of G. J. Holmes, NCSU



Photos by G. Holmes

PM deposition on building



Cathedral of Learning, Pittsburgh, 1930 and 1934. (Figure 10.8 Jacobson, 2012) 55

Effect of acid deposition



Sandstone Figure in 1908 and 1968, Westphalia, Germany
(Figure 10.6 Jacobson, 2012)

Effect of acid deposition



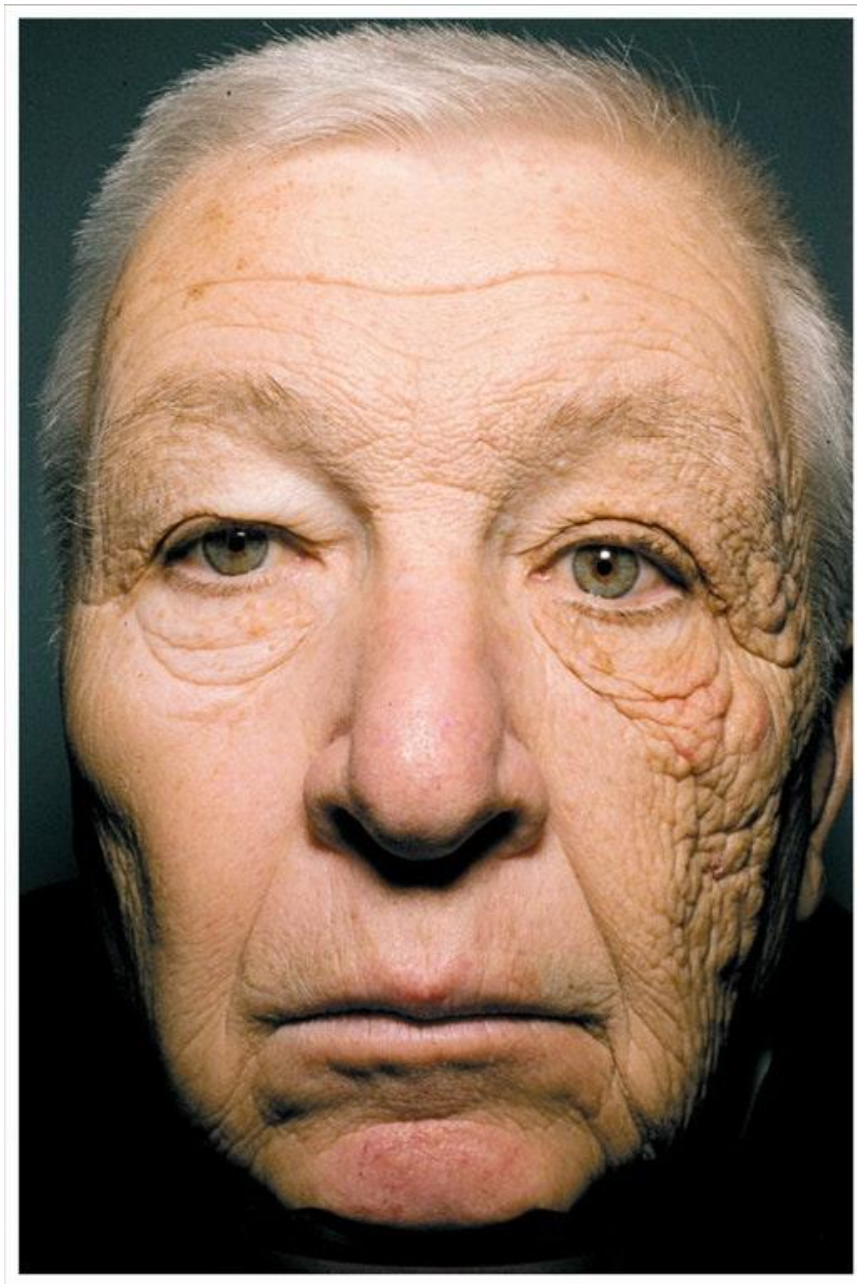
Dead Norway Spruce, Bohemian Forest, Czech Republic, 2005 (www.chmi.cz)

Effects of eutrophication



A boy swims in the algae-filled coastline of Qingdao, China. 2013
(REUTERS/China Daily. <http://zyzc.greatzhonghua.org/archives/14740>)

UV damage skin



Gordon & Brieva. N Engl J Med 2012; 366:e25